

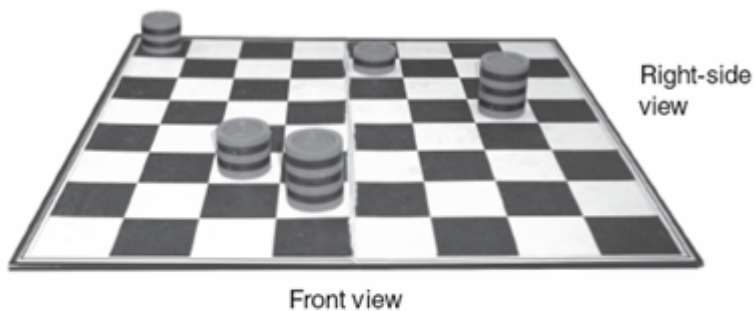


Visualising and creating isometric, orthogonal and oblique drawings

Teacher resource

Question 1

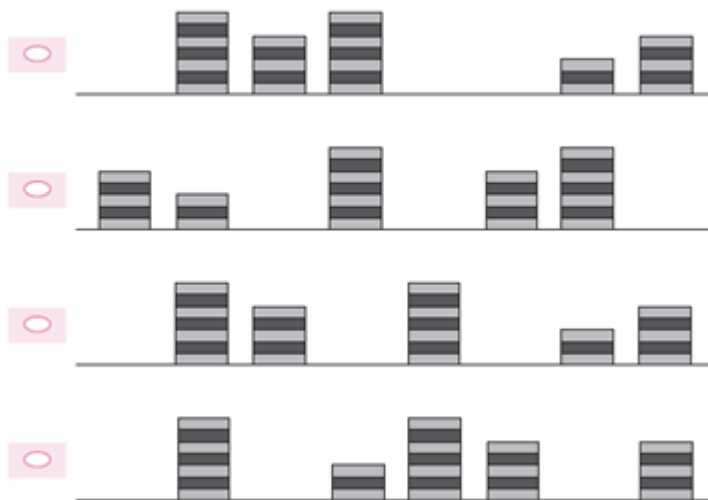
Five stacks of counters have been left on a checkers board.



This is a **Front view**



Which one is the **Right-side view**?



Answer: C

Skill: Students recognise a Right-side view of an object

Background information

Representing a three dimensional object as a drawing in two dimensions is quite challenging for some students. They should understand that while there are a number of techniques to do this - resulting in isometric, oblique, orthogonal or perspective drawings – each type of drawing fulfils a different purpose and each has its limitations. Students should take the purpose, limitations and conventions of each type of drawing into account when creating and interpreting three dimensional drawings. Students will require explicit teaching, discussion and practical experiences to develop these skills and understandings.

In oblique drawings, only the 'front face' is drawn to scale. This means that only the front face in the diagram will accurately depict the angles and relative lengths of the sides of the corresponding face in the original object.

In isometric drawings, the lengths of all edges are to scale and parallelism between sides is maintained in the drawing. The angles in the drawing do not match the corresponding angles in the original object.

Orthogonal drawings show the front, side and top views of the original object as three separate two dimensional figures as they would appear if viewed from that position. There is no attempt to show how the sides are connected to each other.

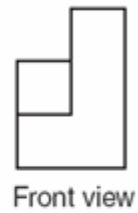
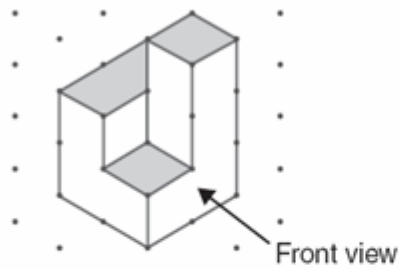
In perspective drawings, the lengths of sides and angles may all be distorted in order to make the drawing look realistic.

Additional questions

1. How will the view from the back compare to the Front-view?
2. How will the Left-side view compare to the Right-side view?

Question 2

This 3D solid is made from 8 cubes glued together.
The front view is drawn.



On the grid below draw the left-side view and the top view of the object.



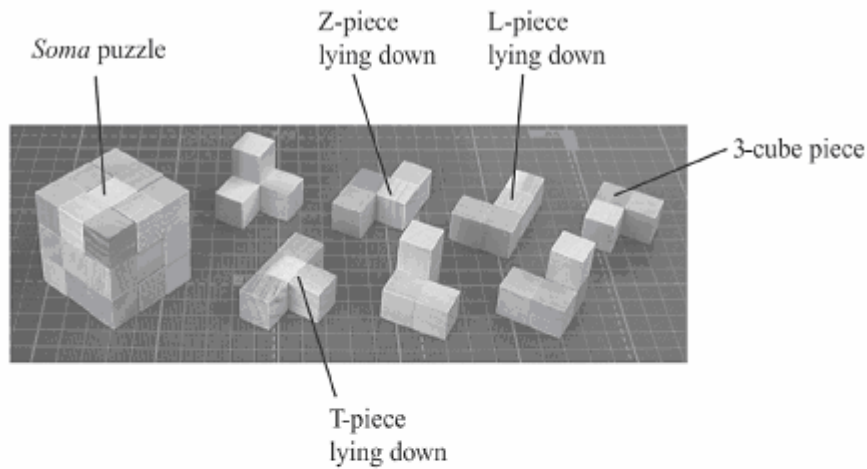
Skill: Students draw orthogonal views of a 3D object given its isometric drawing

Additional questions

1. How many edges does this solid have?
2. If the volume of the solid is 8 cm^3 , what is its surface area?
3. A cube is placed in the step at the front of the solid. Does this change the surface area of the object? Explain your reasoning.

Question 3

The Soma puzzle consists of seven pieces made by joining 3 or 4 wooden cubes together.



Draw the Z-piece in any 'standing-up' position.



Skill: Students draw an isometric drawing of an object made of cubes.

Additional questions

1. Draw the T-piece in a 'standing up' position.
2. Draw the T-piece lying-down from a different view.

Question 4

Which pair of solid objects **cannot** be joined together to make a cube?

Shade one bubble.



Four pairs of solid objects made of cubes are shown, each in a rectangular box. Below each box is a pink oval bubble for shading.

- Pair A: Two identical L-shaped objects, each made of 3 cubes (2 in a row, 1 on top of the left cube).
- Pair B: Two identical L-shaped objects, each made of 3 cubes (2 in a row, 1 on top of the left cube).
- Pair C: One L-shaped object (3 cubes) and one 2x2 square object (4 cubes).
- Pair D: One L-shaped object (3 cubes) and one 2x2 square object (4 cubes).

Answer: D

Skill: Students visualise objects that make a cube.

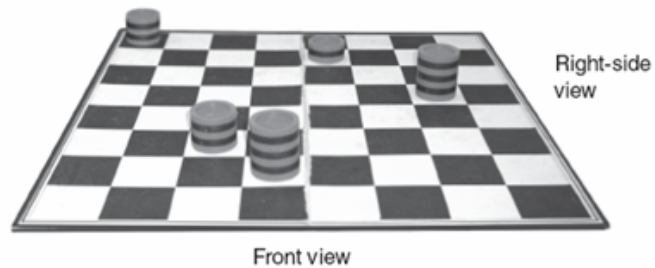
Additional questions

1. What is the surface area of each solid object in pair D?
2. What is the surface area of the cube formed by joining the pair of objects in D?
3. The total of the surface areas of the two solid objects in pair D does not equal the surface area of the cube formed by joining the two objects. Explain why this is so.

Student activity: Visualising and creating isometric, orthogonal and oblique drawings

Question 1

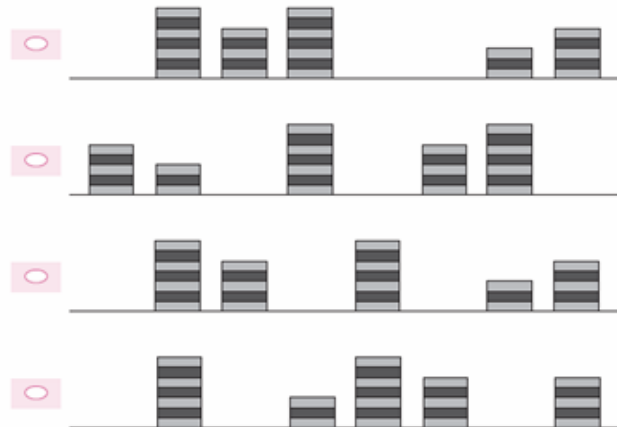
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This is a **Front view**

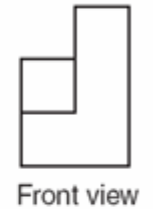
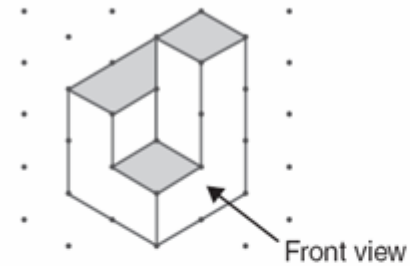


Which one is the **Right-side view**?



Question 2

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The front view is drawn.

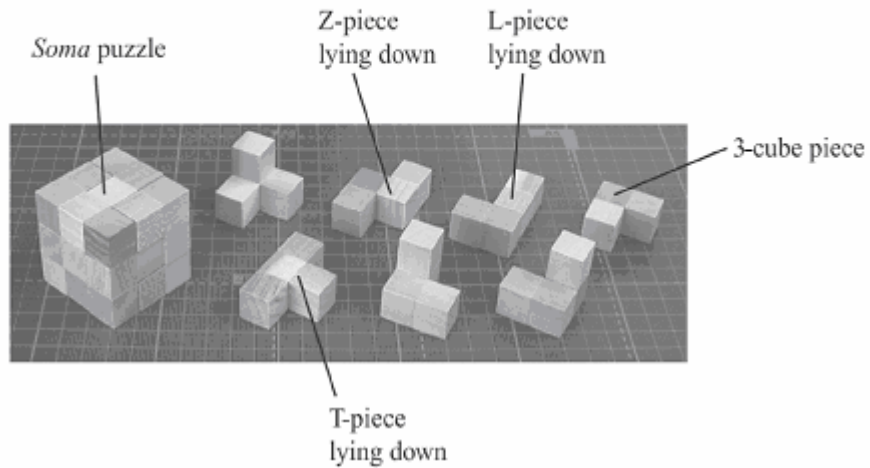


On the grid below draw the left-side view and the top view of the object.



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